

Data Reshaping

Data Wrangling in R

Reshaping: wide vs. long data

<https://github.com/gadenbuie/tidyexplain/blob/main/images/tidyr-pivoting.gif>

wide

id	wide		
	x	y	z
1	a	c	e
2	b	d	f

What is wide/long data?

Data is stored *differently* in the tibble.

Wide: has many columns

```
# A tibble: 1 × 4
  State   June_vacc_rate May_vacc_rate April_vacc_rate
  <chr>         <dbl>         <dbl>         <dbl>
1 Alabama      0.516          0.514          0.511
```

Long: column names become data

```
# A tibble: 3 × 3
  State   name      value
  <chr>   <chr>     <dbl>
1 Alabama June_vacc_rate 0.516
2 Alabama May_vacc_rate  0.514
3 Alabama April_vacc_rate 0.511
```

What is wide/long data?

Wide: multiple columns per individual, values spread across multiple columns

```
# A tibble: 2 × 4
  State      June_vacc_rate May_vacc_rate April_vacc_rate
  <chr>          <dbl>         <dbl>         <dbl>
1 Alabama      0.516          0.514          0.511
2 Alaska       0.627          0.626          0.623
```

Long: multiple rows per observation, a single column contains the values

```
# A tibble: 6 × 3
  State      name      value
  <chr>    <chr>    <dbl>
1 Alabama June_vacc_rate 0.516
2 Alabama May_vacc_rate  0.514
3 Alabama April_vacc_rate 0.511
4 Alaska  June_vacc_rate 0.627
5 Alaska  May_vacc_rate  0.626
6 Alaska  April_vacc_rate 0.623
```

What is wide/long data?

Data is wide or long **with respect** to certain variables.

	Day 1	Day 2	Day 3
Patient 1	A	B	C
Patient 2	D	E	F

Wide

Long

	Day	Value
Patient 1	Day 1	A
Patient 1	Day 2	B
Patient 1	Day 3	C
Patient 2	Day 1	D
Patient 2	Day 2	E
Patient 2	Day 3	F

CC-BY jhudatascience.org

Why do we need to switch between wide/long data?

Wide: **Easier for humans to read**

```
# A tibble: 2 × 4
  State    June_vacc_rate May_vacc_rate April_vacc_rate
  <chr>         <dbl>         <dbl>         <dbl>
1 Alabama      0.516          0.514          0.511
2 Alaska       0.627          0.626          0.623
```

Long: **Easier for R to make plots & do analysis**

```
# A tibble: 6 × 3
  State    name      value
  <chr>   <chr>      <dbl>
1 Alabama June_vacc_rate 0.516
2 Alabama May_vacc_rate  0.514
3 Alabama April_vacc_rate 0.511
4 Alaska  June_vacc_rate 0.627
5 Alaska  May_vacc_rate  0.626
6 Alaska  April_vacc_rate 0.623
```

Pivoting using **tidyr** package

`tidyr` allows you to “tidy” your data. We will be talking about:

- `pivot_longer` - make multiple columns into variables, (wide to long)
- `pivot_wider` - make a variable into multiple columns, (long to wide)
- `separate` - string into multiple columns

The `reshape` command exists. Its arguments are considered more confusing, so we don't recommend it.

You might see old functions `gather` and `spread` when googling. These are older iterations of `pivot_longer` and `pivot_wider`, respectively.

pivot_longer...

Reshaping data from wide to long

`pivot_longer()` - puts column data into rows (tidyr package)

- First describe which columns we want to “pivot_longer”

```
{long_data} <- {wide_data} %>% pivot_longer(cols = {columns to pivot})
```

Reshaping data from wide to long

```
wide_data
```

```
# A tibble: 1 × 3  
  June_vacc_rate May_vacc_rate April_vacc_rate  
    <dbl>         <dbl>         <dbl>  
1      0.516      0.514      0.511
```

```
long_data <- wide_data %>% pivot_longer(cols = everything())  
long_data
```

```
# A tibble: 3 × 2  
  name      value  
  <chr>    <dbl>  
1 June_vacc_rate 0.516  
2 May_vacc_rate  0.514  
3 April_vacc_rate 0.511
```

Reshaping data from wide to long

`pivot_longer()` - puts column data into rows (`tidyr` package)

- First describe which columns we want to “pivot_longer”
- `names_to` = gives a new name to the pivoted columns
- `values_to` = gives a new name to the values that used to be in those columns

```
{long_data} <- {wide_data} %>% pivot_longer(cols = {columns to pivot},  
                                             names_to = {New column name: contains old column names},  
                                             values_to = {New column name: contains cell values})
```

Reshaping data from wide to long

wide_data

```
# A tibble: 1 × 3
  June_vacc_rate May_vacc_rate April_vacc_rate
      <dbl>      <dbl>      <dbl>
1    0.516      0.514      0.511
```

```
long_data <- wide_data %>% pivot_longer(cols = everything(),
                                         names_to = "Month",
                                         values_to = "Rate")
```

long_data

```
# A tibble: 3 × 2
  Month      Rate
  <chr>    <dbl>
1 June_vacc_rate 0.516
2 May_vacc_rate  0.514
3 April_vacc_rate 0.511
```

Newly created column names are enclosed in quotation marks.

Data used: Charm City Circulator

https://sisbid.github.io/Data-Wrangling/data/Charm_City_Circulator_Ridership.csv

```
circ <-  
  read_csv("https://sisbid.github.io/Data-Wrangling/data/Charm_City_Circulator_Ridership.csv")  
head(circ, 5)
```

```
# A tibble: 5 × 15  
  day      date orangeBoardings orangeAlightings orangeAverage purpleBoardings  
  <chr>   <chr>         <dbl>           <dbl>           <dbl>         <dbl>  
1 Monday 01/1...         877             1027             952           NA  
2 Tuesday 01/1...         777             815             796           NA  
3 Wednesday 01/1...    1203            1220            1212.         NA  
4 Thursday 01/1...    1194            1233            1214.         NA  
5 Friday 01/1...    1645            1643            1644           NA  
# i 9 more variables: purpleAlightings <dbl>, purpleAverage <dbl>,  
# greenBoardings <dbl>, greenAlightings <dbl>, greenAverage <dbl>,  
# bannerBoardings <dbl>, bannerAlightings <dbl>, bannerAverage <dbl>,  
# daily <dbl>
```

Reshaping data from wide to long

```
long <- circ %>%  
  pivot_longer(starts_with(c("orange", "purple", "green", "banner")))  
long
```

```
# A tibble: 13,752 × 5
```

	day <chr>	date <chr>	daily <dbl>	name <chr>	value <dbl>
1	Monday	01/11/2010	952	orangeBoardings	877
2	Monday	01/11/2010	952	orangeAlightings	1027
3	Monday	01/11/2010	952	orangeAverage	952
4	Monday	01/11/2010	952	purpleBoardings	NA
5	Monday	01/11/2010	952	purpleAlightings	NA
6	Monday	01/11/2010	952	purpleAverage	NA
7	Monday	01/11/2010	952	greenBoardings	NA
8	Monday	01/11/2010	952	greenAlightings	NA
9	Monday	01/11/2010	952	greenAverage	NA
10	Monday	01/11/2010	952	bannerBoardings	NA

```
# i 13,742 more rows
```

Reshaping data from wide to long

There are many ways to select the columns we want. Use `?tidyr_tidy_select` to look at more column selection options.

```
long <- circ %>%  
  pivot_longer( !c(day, date, daily))  
long
```

```
# A tibble: 13,752 × 5  
   day      date      daily name      value  
   <chr>   <chr>    <dbl> <chr>    <dbl>  
1 Monday 01/11/2010    952 orangeBoardings    877  
2 Monday 01/11/2010    952 orangeAlightings  1027  
3 Monday 01/11/2010    952 orangeAverage      952  
4 Monday 01/11/2010    952 purpleBoardings    NA  
5 Monday 01/11/2010    952 purpleAlightings    NA  
6 Monday 01/11/2010    952 purpleAverage      NA  
7 Monday 01/11/2010    952 greenBoardings     NA  
8 Monday 01/11/2010    952 greenAlightings    NA  
9 Monday 01/11/2010    952 greenAverage      NA  
10 Monday 01/11/2010    952 bannerBoardings    NA  
# i 13,742 more rows
```

Cleaning up long data

We will use `str_replace` from the `stringr` package to put `_` in the names

```
long <- long %>% mutate(
  name = str_replace(name, "Board", "_Board"),
  name = str_replace(name, "Alight", "_Alight"),
  name = str_replace(name, "Average", "_Average")
)
long
```

```
# A tibble: 13,752 × 5
  day   date      daily name          value
<chr> <chr>    <dbl> <chr>          <dbl>
1 Monday 01/11/2010  952 orange_Boardings  877
2 Monday 01/11/2010  952 orange_Alightings 1027
3 Monday 01/11/2010  952 orange_Average    952
4 Monday 01/11/2010  952 purple_Boardings   NA
5 Monday 01/11/2010  952 purple_Alightings  NA
6 Monday 01/11/2010  952 purple_Average     NA
7 Monday 01/11/2010  952 green_Boardings    NA
8 Monday 01/11/2010  952 green_Alightings   NA
9 Monday 01/11/2010  952 green_Average      NA
10 Monday 01/11/2010  952 banner_Boardings   NA
# i 13,742 more rows
```


Cleaning up long data

Now each **var** is Boardings, Averages, or Alightings. We use “**into =**” to name the new columns and “**sep =**” to show where the separation should happen.

```
long <- long %>%  
  separate(name, into = c("line", "type"), sep = "_")  
long
```

```
# A tibble: 13,752 × 6  
  day      date      daily line      type      value  
  <chr>   <chr>    <dbl> <chr>   <chr>    <dbl>  
1 Monday 01/11/2010    952 orange Boardings    877  
2 Monday 01/11/2010    952 orange Alightings  1027  
3 Monday 01/11/2010    952 orange Average     952  
4 Monday 01/11/2010    952 purple Boardings     NA  
5 Monday 01/11/2010    952 purple Alightings    NA  
6 Monday 01/11/2010    952 purple Average     NA  
7 Monday 01/11/2010    952 green  Boardings    NA  
8 Monday 01/11/2010    952 green  Alightings    NA  
9 Monday 01/11/2010    952 green  Average     NA  
10 Monday 01/11/2010    952 banner Boardings    NA  
# i 13,742 more rows
```

GUT CHECK!

What does `pivot_longer()` do?

- A. Summarize data
- B. Subsets data
- C. Reshape data

pivot_wider...

Reshaping data from long to wide

`pivot_wider()` - spreads row data into columns (tidyr package)

- `names_from` = the old column whose contents will be spread into multiple new column names.
- `values_from` = the old column whose contents will fill in the values of those new columns.

```
{wide_data} <- {long_data} %>%  
  pivot_wider(names_from = {Old column name: contains new column names},  
              values_from = {Old column name: contains new cell values})
```

Reshaping data from long to wide

long_data

```
# A tibble: 3 × 2
  Month      Rate
  <chr>    <dbl>
1 June_vacc_rate 0.516
2 May_vacc_rate  0.514
3 April_vacc_rate 0.511
```

```
wide_data <- long_data %>% pivot_wider(names_from = "Month",
                                         values_from = "Rate")
```

wide_data

```
# A tibble: 1 × 3
  June_vacc_rate May_vacc_rate April_vacc_rate
      <dbl>         <dbl>         <dbl>
1      0.516         0.514         0.511
```

Reshaping Charm City Circulator

long

A tibble: 13,752 × 6

	day <chr>	date <chr>	daily <dbl>	line <chr>	type <chr>	value <dbl>
1	Monday	01/11/2010	952	orange	Boardings	877
2	Monday	01/11/2010	952	orange	Alightings	1027
3	Monday	01/11/2010	952	orange	Average	952
4	Monday	01/11/2010	952	purple	Boardings	NA
5	Monday	01/11/2010	952	purple	Alightings	NA
6	Monday	01/11/2010	952	purple	Average	NA
7	Monday	01/11/2010	952	green	Boardings	NA
8	Monday	01/11/2010	952	green	Alightings	NA
9	Monday	01/11/2010	952	green	Average	NA
10	Monday	01/11/2010	952	banner	Boardings	NA

i 13,742 more rows

Reshaping Charm City Circulator

```
wide <- long %>% pivot_wider(names_from = "type",  
                             values_from = "value")  
wide
```

```
# A tibble: 4,584 × 7
```

	day <chr>	date <chr>	daily <dbl>	line <chr>	Boardings <dbl>	Alightings <dbl>	Average <dbl>
1	Monday	01/11/2010	952	orange	877	1027	952
2	Monday	01/11/2010	952	purple	NA	NA	NA
3	Monday	01/11/2010	952	green	NA	NA	NA
4	Monday	01/11/2010	952	banner	NA	NA	NA
5	Tuesday	01/12/2010	796	orange	777	815	796
6	Tuesday	01/12/2010	796	purple	NA	NA	NA
7	Tuesday	01/12/2010	796	green	NA	NA	NA
8	Tuesday	01/12/2010	796	banner	NA	NA	NA
9	Wednesday	01/13/2010	1212.	orange	1203	1220	1212.
10	Wednesday	01/13/2010	1212.	purple	NA	NA	NA

```
# i 4,574 more rows
```

Adding prefixes

Prefixes when pivoting

the `datasets::airquality` data shows various air quality metrics measured in New York in 1973.

```
air <- datasets::airquality %>% select(Temp, Month, Day)  
air
```

	Temp	Month	Day
1	67	5	1
2	72	5	2
3	74	5	3
4	62	5	4
5	56	5	5
6	66	5	6
7	65	5	7
8	59	5	8
9	61	5	9
10	69	5	10
11	74	5	11
12	69	5	12
13	66	5	13
14	68	5	14
15	58	5	15
16	64	5	16
17	66	5	17
18	57	5	18
19	68	5	19
20	62	5	20
21	59	5	21

Prefixes when pivoting

Let's pivot **Month** wider: but it might be helpful to add "Month" to the new column name so it isn't just numbers.

```
air %>% pivot_wider(names_from = "Month",  
                    values_from = "Temp")
```

```
# A tibble: 31 × 6  
  Day `5` `6` `7` `8` `9`  
  <int> <int> <int> <int> <int> <int>  
1     1    67    78    84    81    91  
2     2    72    74    85    81    92  
3     3    74    67    81    82    93  
4     4    62    84    84    86    93  
5     5    56    85    83    85    87  
6     6    66    79    83    87    84  
7     7    65    82    88    89    80  
8     8    59    87    92    90    78  
9     9    61    90    92    90    75  
10    10    69    87    89    92    73  
# i 21 more rows
```

Prefixes when pivoting

Much better!

```
air %>% pivot_wider(names_from = "Month",  
                    values_from = "Temp",  
                    names_prefix = "Month_")
```

```
# A tibble: 31 × 6  
  Day Month_5 Month_6 Month_7 Month_8 Month_9  
  <int>   <int>   <int>   <int>   <int>   <int>  
1     1     67     78     84     81     91  
2     2     72     74     85     81     92  
3     3     74     67     81     82     93  
4     4     62     84     84     86     93  
5     5     56     85     83     85     87  
6     6     66     79     83     87     84  
7     7     65     82     88     89     80  
8     8     59     87     92     90     78  
9     9     61     90     92     90     75  
10    10     69     87     89     92     73  
# i 21 more rows
```

Summary

- `tidyr` package helps us convert between wide and long data
- `pivot_longer()` goes from wide -> long
 - Specify columns you want to pivot
 - Specify `names_to =` and `values_to =` for custom naming
- `pivot_wider()` goes from long -> wide
 - Specify `names_from =` and `values_from =`

<https://sisbid.github.io/Data-Wrangling/labs/data-resaping-lab.Rmd>